

Linear systems and elimination

Sharp inverse norm bound from upper bounds on matrix entries

Difficulty: challenging **Importance:** interesting to specialist

Status: Solved

Topic: Conditioning of positive diagonally dominant linear systems.

Independently reviewed resolution - 2026-09-11

Negative resolution and sharp replacement. Section 1 gives an admissible positive symmetric strictly diagonally dominant 3×3 matrix with inverse infinity norm $7/9$, below the proposed comparison value $5/4$. Theorem 1 in Section 2 proves that the exact infimum over the displayed class is $1/(\alpha + m)$ for every allowed parameter choice, and strict positivity prevents attainment. The order is entrywise; stronger comparisons of dominance margins are outside the result.

Author: Matthew J. Colbrook, Department of Applied Mathematics and Theoretical Physics, University of Cambridge. [Complete manuscript](#), [independent proof review](#), and [submission record](#). The supplied notes were reconstructed with substantial AI assistance. This is independent agent verification, not external human peer review or formal proof-assistant certification; no novelty or priority claim is made.

The difficulty, importance and rating rationale below are historical assessments of the original open target. Original statements, references and dated audits are preserved.

Rating rationale: Challenging reflects a sharp inverse-norm inequality over a constrained matrix family; specialist impact concerns extremal conditioning for positive diagonally dominant systems.

Problem statement

Let $n \geq 3$, $m > 0$, and $\alpha \geq (n - 2)m$, and define

$$S = \alpha I_n + m \mathbf{1} \mathbf{1}^T, \quad \mathbf{1} = (1, \dots, 1)^T \in \mathbb{R}^n.$$

For every real symmetric matrix J satisfying

$$0 < J_{ij} \leq S_{ij} \quad (1 \leq i, j \leq n), \quad J_{ii} \geq \sum_{j \neq i} J_{ij} \quad (1 \leq i \leq n),$$

does the sharp bound

$$\|J^{-1}\|_{\infty} \geq \frac{\alpha + 2m(n - 1)}{\alpha(\alpha + mn)} = \|S^{-1}\|_{\infty}$$

hold, with equality if and only if $J = S$? Here $\|M\|_{\infty} = \max_i \sum_j |M_{ij}|$. The hypotheses make J nonsingular. The inequalities on entries are entrywise, not Loewner inequalities.

Why it matters

This would give an optimal lower bound on inverse amplification for a structured family of linear systems using readily available entry bounds.

References

- C. J. Hillar, S. Lin, and A. Wibisono, [Inverses of symmetric, diagonally dominant positive matrices and applications](#), §8, Conjecture 8.1, p. 17; the opening notation defines entrywise ordering. The parameter range for S is inherited from Theorem 6.1 and the paragraph preceding Conjecture 8.1.
- C. J. Hillar and A. Wibisono, [A Hadamard-type lower bound for symmetric diagonally dominant positive matrices](#), *Linear Algebra and its Applications* 472 (2015), 135–141, §3, Conjecture 3.4 and Theorem 3.5. This follow-up settles the separate determinant conjecture, not the inverse norm claim.

Earlier status check — 2026-09-08

On 2026-09-08, searched the original title and combinations of “Hillar”, “Lin”, “Wibisono”, “Conjecture 8.1”, “Conjecture 7.1”, “Tight bounds on the infinity norm”, and “proof”. The same inverse claim also occurs as Conjecture 7.1 in the authors’ revised manuscript titled *Tight bounds on the infinity norm of inverses of symmetric diagonally dominant positive matrices*. No solution was located. The 2015 determinant resolution was checked separately. No recent explicit reaffirmation was found. The displayed parameter restriction makes explicit the diagonally dominant comparison matrix used by the source; it does not replace the conjecture’s entrywise comparison by the stronger bound on diagonal dominance margins mentioned before it.

Audit update — 2026-09-10

Rechecked [Hillar–Lin–Wibisono](#), §8, Conjecture 8.1: the extremizer and entrywise hypothesis agree with this statement. Searches for that conjecture and later inverse bounds found no proof of the displayed sharp inequality. Later determinant results cited above concern a different objective. The open verdict relies on this historical explicit conjecture and the bounded follow-up search.